STE General Workflow

Generates a comprehensive elephant movement analysis report for Save the Elephants, combining GPS observations from EarthRanger with Google Earth Engine satellite imagery into home range, habitat use, activity, and event maps, charts, and reports.

What it produces

The workflow processes GPS observations for a chosen subject group and produces a full suite of spatial, temporal, and ecological analyses:

Movement maps

- A **Movement Tracks Map** — current and previous period trajectories overlaid together, coloured by period (current vs. previous) for direct comparison
- A **Subject Tracks Map** — trajectories coloured per individual subject using EarthRanger hex colours

Home range maps

- An **Overall Home Range Map** — Elliptical Time Density (ETD) at percentiles 50–99.9%, coloured by the RdYlGn palette
- A **Filtered Home Range Map** — 99th percentile home range only, shown in a single chocolate tone
- A **Protected Areas Home Range Map** — ETD computed from fixes inside protected land (Community Conservancies, National Reserves, National Parks)
- An **Unprotected Areas Home Range Map** — ETD computed from fixes outside protected land
- **Seasonal Home Range Maps** (per subject) — 99.9th percentile ETD split into wet and dry seasons, coloured by season

Habitat use maps

- A **Recursion Events Raster** — ecograph network weight raster classified into 6 natural-break bins, green → red
- A **Dry Season Mean Speed Raster** — mean movement speed raster for dry-season fixes, 6 natural-break bins, green → red
- A **Wet Season Mean Speed Raster** — same as above for wet-season fixes

Activity maps

- A **Time of Day Dominance Map** — grid showing whether each cell is used predominantly at night (dark blue) or during the day (cornflower blue)

- A **Night Fixes Map** — grid cells where ≥ 65% of fixes are nocturnal

Event maps

- An **Elephant Sightings / Collaring Events Map** — scatter plot of EarthRanger events for the analysis period

Charts and tables

- A **Protection Status Bar Chart** — proportion of fixes inside vs. outside protected areas
- A **Seasonal Windows CSV** — NDVI-derived wet/dry season date ranges used to label trajectories

Reports

- A **Word document report** (`.docx`) — cover page (logo, subject count, report period) plus a general per-subject section
- An **interactive dashboard**

Requirements

- Access to an **EarthRanger** instance containing the subject group to analyse
- Access to a **Google Earth Engine** service account (for NDVI-based seasonal window detection and satellite base layers)
- No separate landDx file is needed to get started — the workflow downloads the standard LandDx protected-areas layer automatically for the default **Map Overlay** option

---

1. Load the Workflow

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste this repository's URL into the **Github Link** field, then click **Add Template**:

```
https://github.com/wildlife-dynamics/ste-general.git
```

Once added, it appears in the **Workflow Templates** list as **ste_general**. Click it to open the workflow configuration form (the card may show **Initializing...** briefly while the environment is set up).

---

2. Configure the Workflow

You'll be prompted to fill in the following parameters before running.

Data Source Connections

Navigate to **Data Sources** and click **Connect**. A dialog will prompt you to **Select Data Source Type**. This workflow requires two connections:

Data Source Type	Purpose
EarthRanger	Pull subject observations and collaring events
Google Earth Engine	Fetch satellite imagery and compute NDVI-based seasonal windows

EarthRanger connection — select **EarthRanger**, then fill in:

- **Data Source Name** — a label to identify this connection
- **EarthRanger URL** — your instance URL (e.g. `your-site.pamdas.org`)
- **EarthRanger Username** and **EarthRanger Password**

> Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

Google Earth Engine connection — select **Google Earth Engine**, then fill in:

- **Data Source Name** — a label to identify this connection
- **Private Key** — click **Browse** to select your GEE service account private key file (JSON format)

> To generate a private key, follow the instructions in the Setup Guide. The key is stored encrypted and used only to authenticate with Google Earth Engine.

Once both connections exist, select them from the **Connect to earth ranger** and **Connect to earth engine** dropdowns in the workflow form.

Workflow Details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes (e.g. date range or subject group)

Basemap Layers

The **Configure basemap layers** step sets the two stacked background tile layers shared by all maps. Pre-filled with sensible defaults, but the URL, opacity, and max zoom of each layer are editable.

Layer	Default Opacity	Max Zoom
ESRI World Hillshade	`0.995`	`20`
ESRI World Boundaries & Places	`0.35`	`20`

Analysis Time Range

Field	Description
Timezone	Select the local timezone (e.g. `Africa/Nairobi UTC+03:00`)
Since	Start date and time of the analysis period
Until	End date and time of the analysis period

All subject tracks, home ranges, and activity maps are computed within this window.

Previous Period Range

This period drives the "Previous tracks" layer on the Movement Tracks Map, enabling direct before-and-after comparison alongside the current period tracks. Every mode computes a comparison period that ends on your time range's Start Date, then works backward to determine its own Start Date.

Offset Mode	Default	Description
Preset	`1 month back`	A common lookback: `Same as current period`, `1 month back`, `3 months back`, `6 months back`, or `1 year back`
Calendar	—	Pick an exact Start Date for the comparison period
Custom	Years `0`, Months `1`, Weeks `0`, Days `0`	Manually set how far back the comparison period's Start Date should be

Subject Group

Field	Default	Description
Subject Group Name	`Elephants`	Name of the subject group to pull from EarthRanger (case-sensitive)

Map Overlay

Choose the overlay layer displayed on every map. This overlay is also used to calculate which GPS fixes fall inside vs. outside protected land, which drives the Protected/Unprotected Home Range maps and the Protection Status Bar Chart. Only protected-area style layers are supported for now.

Option	When to use	What to provide
LandDx (Default Protected Areas) — default	Standard use case	Nothing — the workflow automatically downloads and styles the standard LandDx GeoPackage. It is filtered to three area types and coloured as follows: Community Conservancy `#a6b697` (sage green), National Reserve `#88a78e` (muted green), National Park `#115631` (dark forest green)
EarthRanger Spatial Feature	Your protected areas are maintained as EarthRanger spatial features	One or more layers, each specifying what to load (featureset / type / id) and optional style overrides
Custom Local File	You have a local GeoJSON / GeoPackage / GeoParquet file	One or more local files (or split groups within a file), each becoming its own styled layer

Trajectory Segment Filter

Applied to both the current and previous period trajectories before any analysis. Removes GPS noise and biologically unrealistic movement segments.

Field	Default	Description
Minimum Segment Length (m)	`0.001`	Discard segments shorter than this distance
Maximum Segment Length (m)	`5000`	Discard segments longer than this distance
Minimum Segment Duration (s)	`1`	Discard segments shorter than this duration
Maximum Segment Duration (s)	`21600`	Discard segments longer than this duration (6 hours)
Minimum Speed (km/h)	`0.01`	Discard segments where average speed falls below this value
Maximum Speed (km/h)	`9`	Discard segments where average speed exceeds this value (a sensible ceiling for elephants)

Map Zoom & Extent

Calculates the geographic extent of the movement tracks and determines an appropriate zoom level and center point, ensuring all tracks are visible and well-framed.

Field	Default	Description
Expansion Factor	`1.5`	Factor to expand the bounding box around the tracks (e.g. `1.2` = 20% larger)

Retrieve All Events

Field	Default	Description
Event Types	*(empty)*	Filter to specific EarthRanger event types (click **Add** to add entries; leave empty to include all event types)
Force Point Geometry	`true`	If enabled, polygon/multipolygon event geometries are reduced to their centroid; disable to preserve native geometry types

Report Logo

Optional — provide the logo image to appear on the report cover page. Leave blank to omit a logo.

Input Method	What to provide
Download from URL	Paste the direct download URL into the **URL** field
Use local file	Enter the absolute path to the logo file on the machine running the workflow

Raster Maps

Configures how the mean speed raster is generated from movement trajectories.

Field	Default	Description
Step Length / Grid Size (m)	`2000`	Pixel size (in meters) used to generate the Recursion Events raster **and** the dry/wet season Mean Speed rasters — one value drives all three. Smaller values are more detailed but slower to generate and produce larger files

3. Run the Workflow

Once all parameters are configured, click **Submit**. The runner will:

1. Pull subject GPS observations (current and previous period) from EarthRanger.
2. Process relocations and apply the trajectory segment filter to both periods.
3. Query Google Earth Engine to determine NDVI-derived wet and dry seasonal windows; label each trajectory segment accordingly.
4. Load and style the selected Map Overlay layer (LandDx protected areas by default).
5. Generate the Movement Tracks Map (current + previous overlay) and the Subject Tracks Map.
6. Fetch EarthRanger events and generate the Elephant Sightings map.
7. Compute Elliptical Time Density home ranges: overall, filtered 99th percentile, protected/unprotected split, and per-subject seasonal.
8. Generate the Recursion Events raster and dry/wet season Mean Speed rasters.
9. Compute the Day/Night Dominance grid and Night Fixes grid.
10. Render the Protection Status bar chart.
11. Assemble the Word report (cover page + general section) and the dashboard.
12. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS`.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/`:

File	Description
`relocations.geoparquet`	Current period GPS fix locations with day/night labels
`previous_period_relocations.geoparquet`	Previous period GPS fix locations
`trajectories.geoparquet`	Current period trajectory segments with season labels
`previous_period_trajectories.geoparquet`	Previous period trajectory segments
`homerange_etd.geoparquet`	Overall ETD home range polygons (50–99.9th percentiles)
`homerange_protected_etd.geoparquet`	ETD home range for fixes inside protected areas
`homerange_unprotected_etd.geoparquet`	ETD home range for fixes outside protected areas
`<subject>_seasonal_etd.geoparquet`	Seasonal ETD per subject
`seasonal_windows.csv`	NDVI-derived wet/dry season date ranges
`weighted_raster.*`	Recursion events ecograph raster file
`movement_tracks.html`	Interactive movement tracks map (current + previous period)
`subject_tracks.html`	Interactive subject tracks map (coloured per individual)

File	Description
`collared_points.html`	Interactive elephant sightings / collaring events map
`overall_homerange.html`	Interactive overall home range map
`filtered_homerange.html`	Interactive 99th percentile home range map
`protected_areas.html`	Interactive protected areas home range map
`unprotected_areas.html`	Interactive unprotected areas home range map
`<subject>_seasonal_homerange.html`	Interactive seasonal home range map per subject
`recursion_events.html`	Interactive recursion events raster map
`dry_mean_speed_raster_map.html`	Interactive dry season mean speed raster map
`wet_mean_speed_raster_map.html`	Interactive wet season mean speed raster map
`time_of_day_dominance.html`	Interactive time of day dominance grid map
`night_fixes.html`	Interactive night fixes grid map
`protection_status_bar.html`	Interactive protection status bar chart
`movement_tracks.png`	Screenshot of movement tracks map (2× resolution)
`subject_tracks.png`	Screenshot of subject tracks map (2× resolution)
`collared_points.png`	Screenshot of collaring events map (2× resolution)
`overall_homerange.png`	Screenshot of overall home range map (2× resolution)
`filtered_homerange.png`	Screenshot of filtered 99th percentile home range map (2× resolution)
`protected_areas.png`	Screenshot of protected areas home range map (2× resolution)
`unprotected_areas.png`	Screenshot of unprotected areas home range map (2× resolution)
`<subject>_seasonal_homerange.png`	Screenshot of seasonal home range per subject (2× resolution)
`recursion_events.png`	Screenshot of recursion events raster map (2× resolution)
`dry_mean_speed_raster_map.png`	Screenshot of dry season speed raster map (2× resolution)

File	Description
`wet_mean_speed_raster_map.png`	Screenshot of wet season speed raster map (2× resolution)
`time_of_day_dominance.png`	Screenshot of time of day dominance map (2× resolution)
`night_fixes.png`	Screenshot of night fixes map (2× resolution)
`protection_status_bar.png`	Screenshot of protection status bar chart (2× resolution)
`general_cover_page.docx`	Rendered report cover page (logo, subject count, report period)
`overall_report.docx`	Final combined Word report (cover page + general per-subject section)

More Help

- **Technical Guide:** technical_guide/ste_general_technical_guide.pdf — pipeline internals and calculation methodology
- **Issues:** github.com/wildlife-dynamics/ste-general/issues

Development

This workflow's code (`ecoscope-workflows-ste-general-workflow/`) is generated from `spec.yaml` and `test-cases.yaml` . After editing either file, recompile and commit the generated changes:

```
pixi run --manifest-path pixi.toml --locked bash -c "./dev/recompile.sh --update"
```

License

BSD 3-Clause